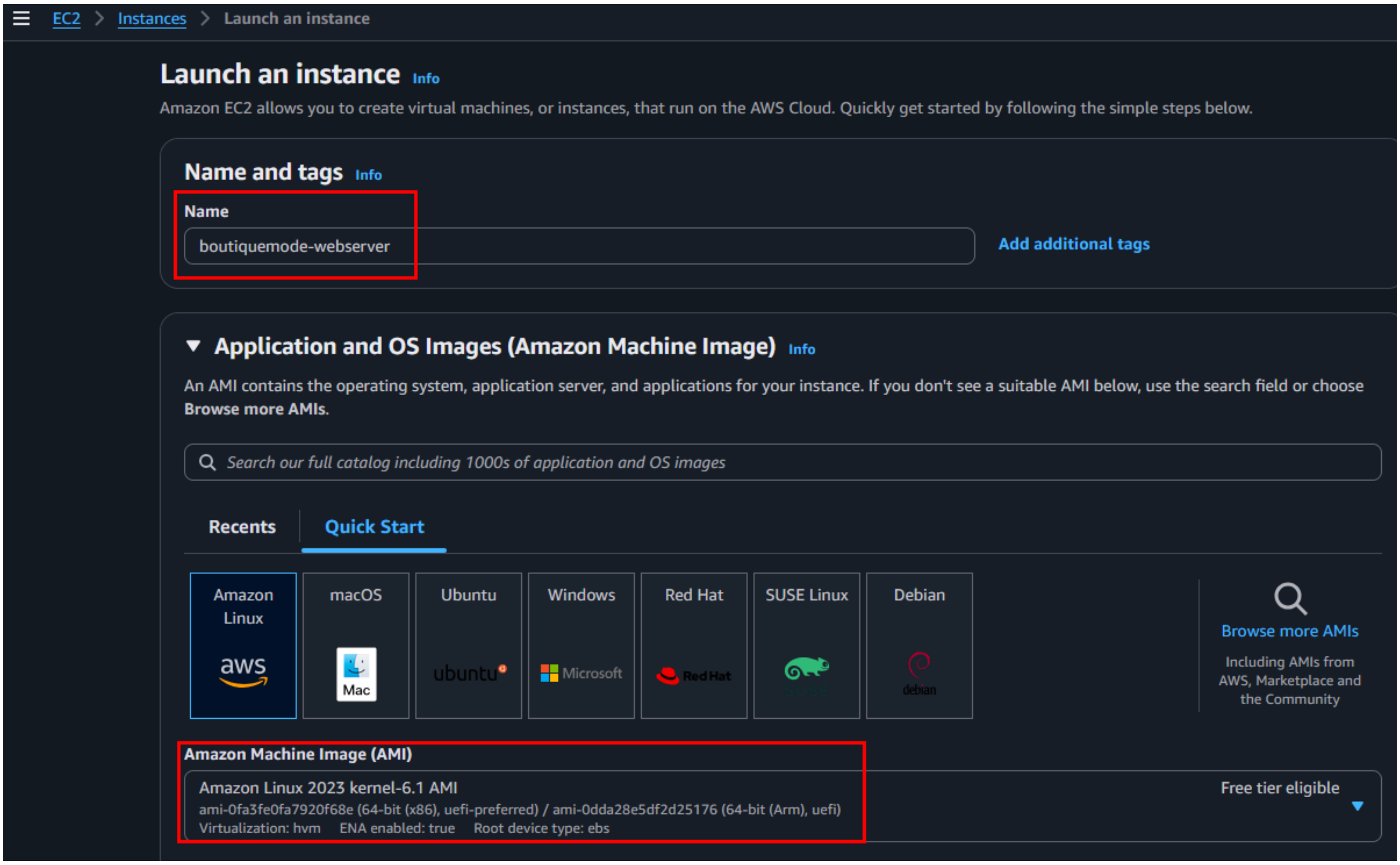
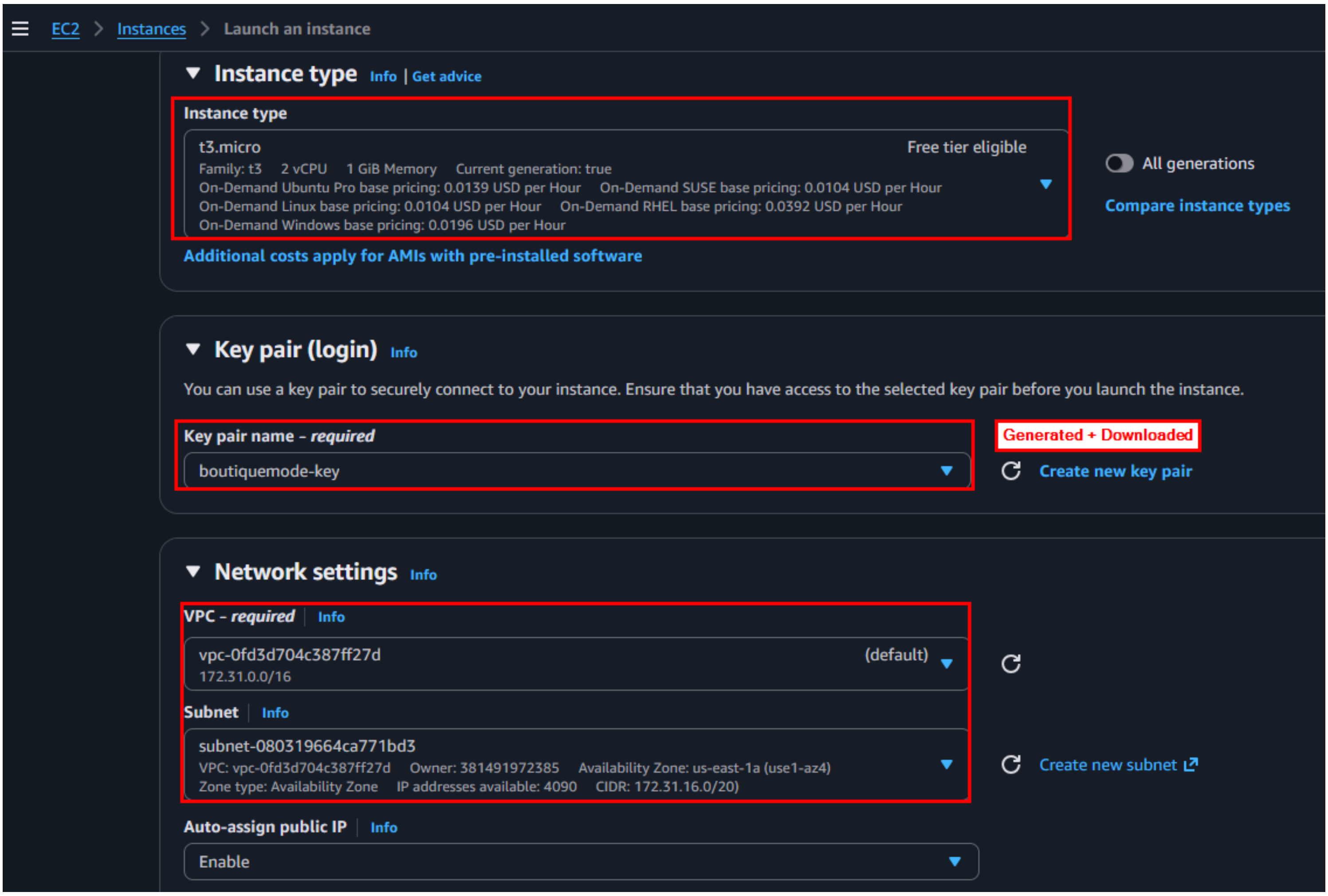
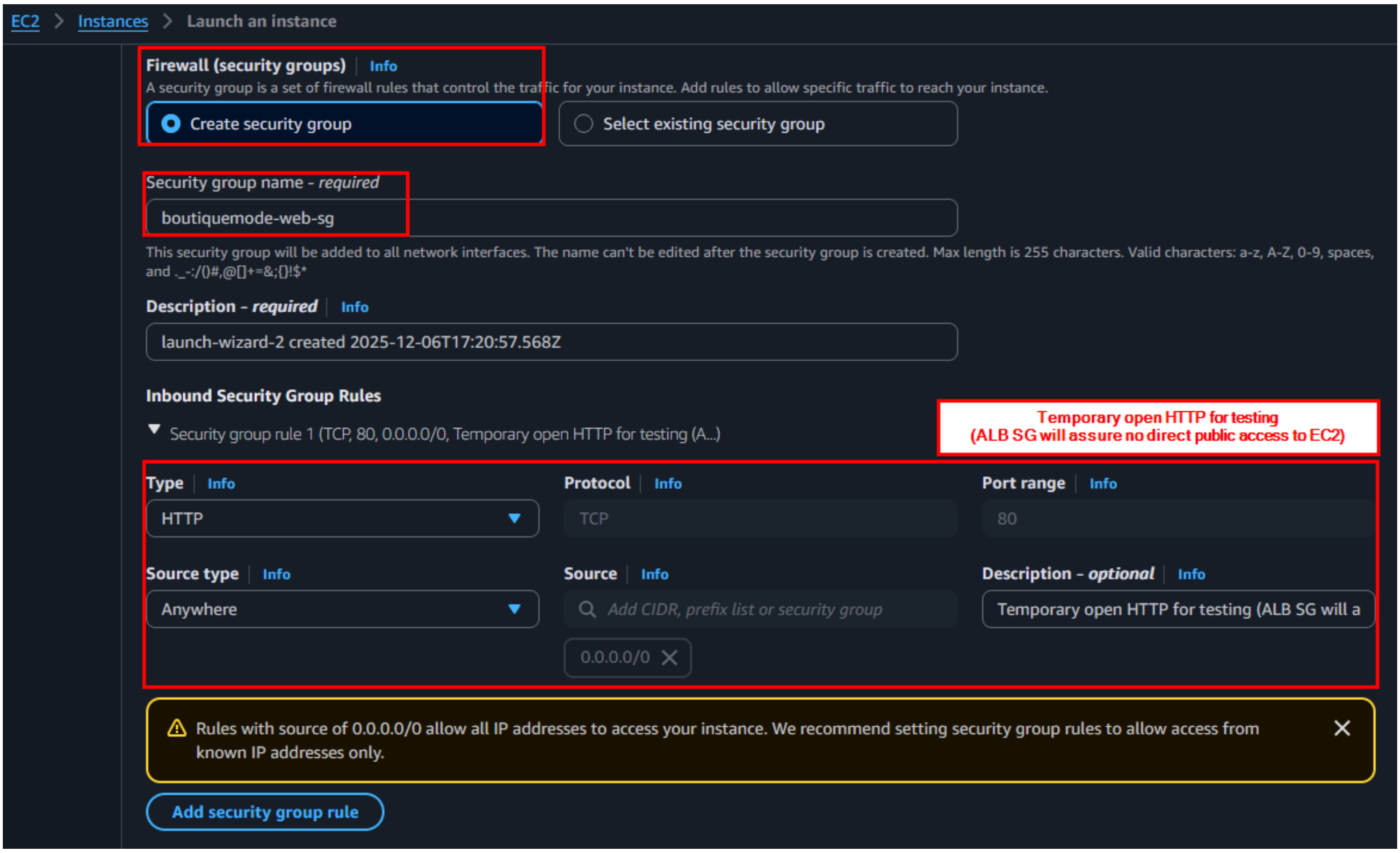
Phase 2: EC2 instance and install + configure Nginx

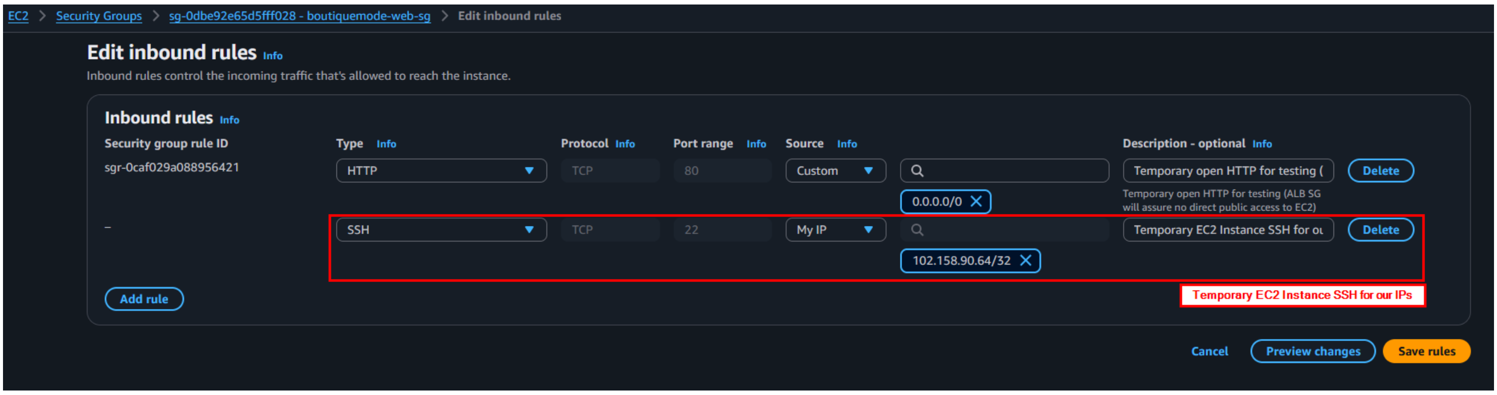
Step 1: Launch the Instance



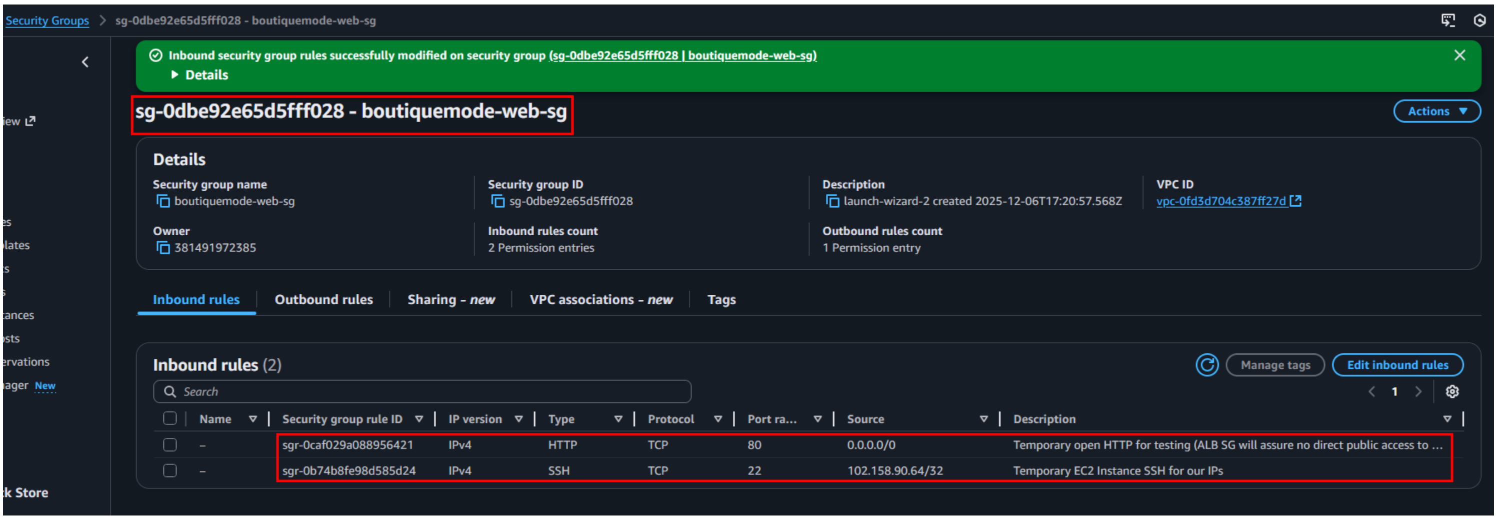

EC2 > Instances > Launch an instance
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
boutiquemode-key
Generated + Downloaded
Create new key pair
Network settings Info
VPC - required Info
vpc-0fd3d704c387ff27d
(default)
172.31.0.0/16
Subnet Info
subnet-080319664ca771bd3
VPC: vpc-0fd3d704c387ff27d Owner: 381491972385 Availability Zone: us-east-1a (use1-az4)
Zone type: Availability Zone IP addresses available: 4090 CIDR: 172.31.16.0/20)
Create new subnet
Auto-assign public IP Info
Enable


EC2 > Instances > Launch an instance
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
boutiquemode-web-sg
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-2 created 2025-12-06T17:20:57.568Z
Inbound Security Group Rules
Security group rule 1 (TCP, 80, 0.0.0.0/0, Temporary open HTTP for testing (A...)
Temporary open HTTP for testing
(ALB SG will assure no direct public access to EC2)
Type Info
HTTP
Protocol Info
TCP
Port range Info
80
Source type Info
Anywhere
Source Info
Add CIDR, prefix list or security group
0.0.0.0/0
Description - optional Info
Temporary open HTTP for testing (ALB SG will a
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Add security group rule

SSH Access:

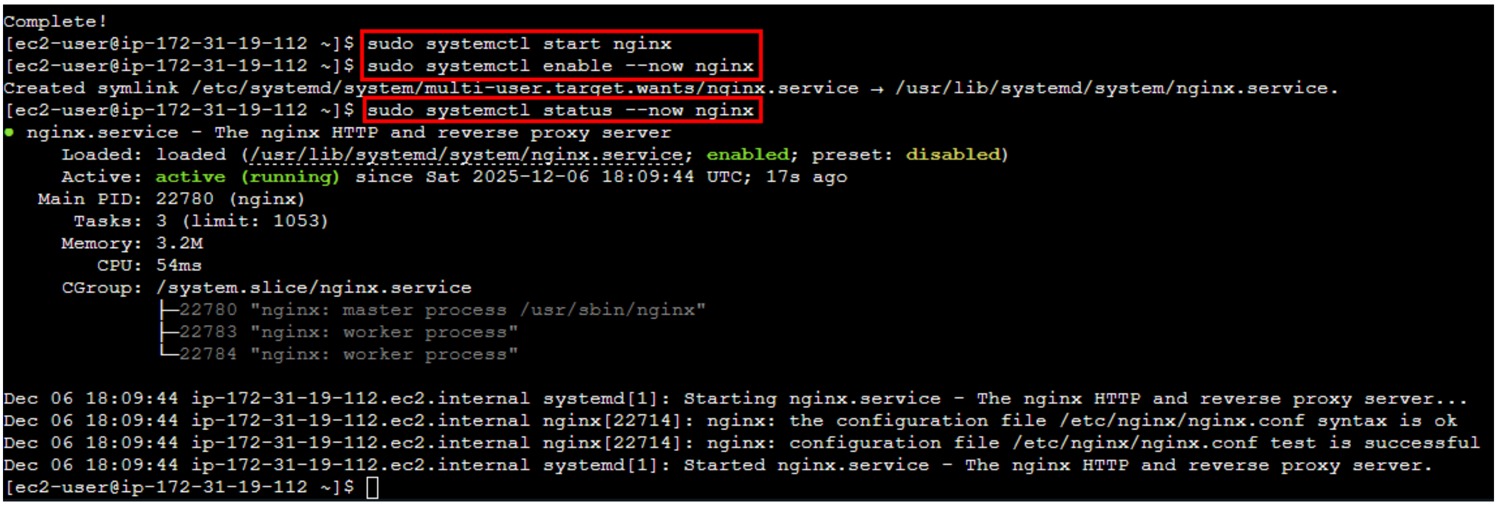


Verification:



Step 2: NGINX Web Server Setup

```
       #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-172-31-19-112 ~]$ sudo yum update -y

Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-19-112 ~]$ sudo yum install nginx -y
Last metadata expiration check: 0:00:12 ago on Sat Dec  6 18:09:19 2025.
Dependencies resolved.
================================================================================
 Package                         Architecture          Version
================================================================================
Installing:
 nginx                           x86_64                1:1.28.0-1.amzn2023.0.2
Installing dependencies:
 generic-logos-httpd             noarch                18.0.0-12.amzn2023.0.3
 gperftools-libs                 x86_64                2.9.1-1.amzn2023.0.3
 libunwind                       x86_64                1.4.0-5.amzn2023.0.3
 nginx-core                      x86_64                1:1.28.0-1.amzn2023.0.2
 nginx-filesystem                noarch                1:1.28.0-1.amzn2023.0.2
 nginx-mimetypes                 noarch                2.1.49-3.amzn2023.0.3

Transaction Summary
================================================================================
Install  7 Packages
```



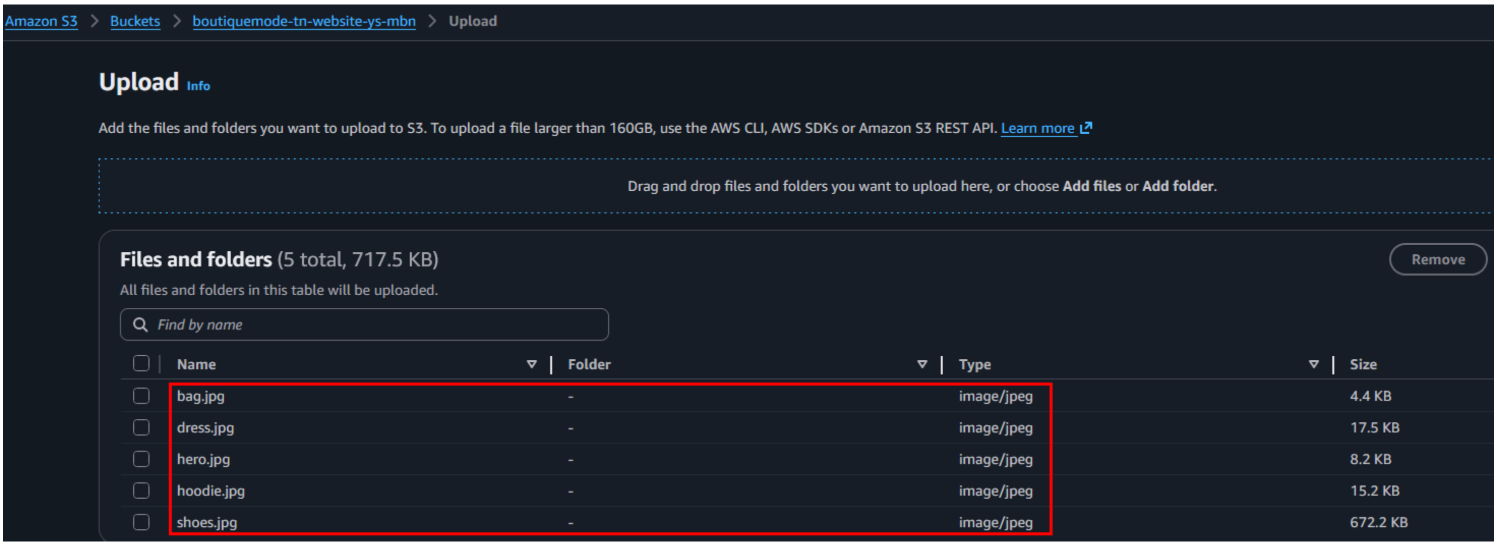
```
Complete!
[ec2-user@ip-172-31-19-112 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-19-112 ~]$ sudo systemctl enable --now nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-172-31-19-112 ~]$ sudo systemctl status --now nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Sat 2025-12-06 18:09:44 UTC; 17s ago
   Main PID: 22780 (nginx)
      Tasks: 3 (limit: 1053)
     Memory: 3.2M
        CPU: 54ms
     CGroup: /system.slice/nginx.service
             ├─22780 "nginx: master process /usr/sbin/nginx"
             ├─22783 "nginx: worker process"
             └─22784 "nginx: worker process"

Dec 06 18:09:44 ip-172-31-19-112.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Dec 06 18:09:44 ip-172-31-19-112.ec2.internal nginx[22714]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Dec 06 18:09:44 ip-172-31-19-112.ec2.internal nginx[22714]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Dec 06 18:09:44 ip-172-31-19-112.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-19-112 ~]$
```

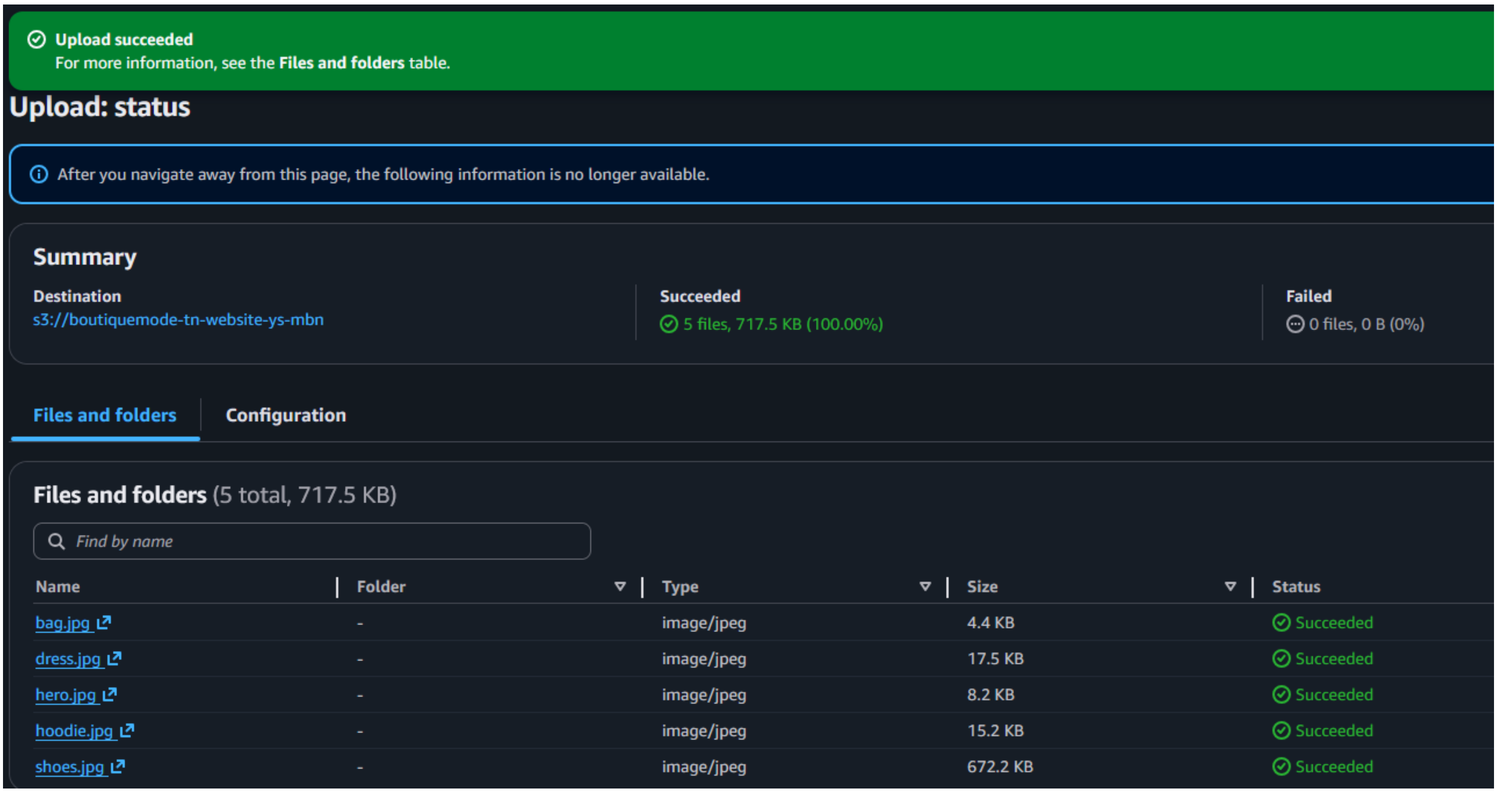
```
# sudo yum update -y
# sudo yum install nginx -y
# sudo systemctl start nginx
# sudo systemctl enable nginx

# curl http://localhost

# sudo rm -f /usr/share/nginx/html/index.html
# sudo bash -c 'cat > /usr/share/nginx/html/index.html <<EOF
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>BoutiqueMode.tn – Fashion Redefined</title>
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <style>
        body {
            margin: 0; font-family: Arial, sans-serif; background: #f5f5f5;
        }
        header {
            background-color: #111; color: white; padding: 20px; text-align: center;
        }
        nav {
            background: #222; padding: 10px; text-align: center;
        }
        nav a {
            color: white; text-decoration: none; margin: 0 15px; font-weight: bold;
        }
        .hero {
            background: url("hero.jpg") center/cover no-repeat;
```

```
            height: 350px;
            display: flex;
            align-items: center;
            justify-content: center;
            color: white;
            font-size: 2rem;
            text-shadow: 1px 1px 4px black;
        }
        .grid {
            display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
            gap: 20px; padding: 30px;
        }
        .card {
            background: white; border-radius: 10px; box-shadow: 0 2px 6px rgba(0,0,0,0.1);
            overflow: hidden; text-align: center; padding-bottom: 20px;
        }
        .card img {
            width: 100%; height: 220px; object-fit: cover;
        }
        footer {
            background: #111; color: white; text-align: center; padding: 20px; margin-top:
40px;
        }
    </style>
</head>
<body>

<header>
    <h1>BoutiqueMode.tn</h1>
    <p>Your Online Fashion Destination</p>
</header>

<nav>
    <a href="#">Home</a>
    <a href="#">Women</a>
    <a href="#">Men</a>
    <a href="#">Accessories</a>
    <a href="#">Contact</a>
</nav>

<div class="hero">
    Discover the New Collection 2025 ✨
```

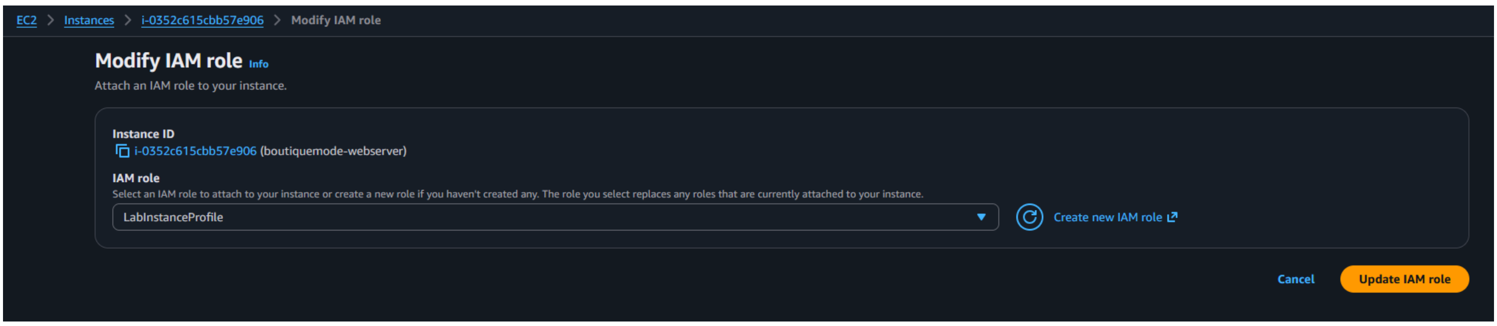
```
</div>

<section class="grid">
   <div class="card">
      <img src="dress.jpg" alt="Dress" onerror="this.src='https://via.placeholder.com/300'">
      <h3>Elegant Dress</h3>
      <p>120 TND</p>
   </div>

   <div class="card">
      <img src="shoes.jpg" alt="Shoes" onerror="this.src='https://via.placeholder.com/300'">
      <h3>Premium Shoes</h3>
      <p>189 TND</p>
   </div>

   <div class="card">
      <img src="bag.jpg" alt="Bag" onerror="this.src='https://via.placeholder.com/300'">
      <h3>Designer Bag</h3>
      <p>149 TND</p>
   </div>

   <div class="card">
      <img src="hoodie.jpg" alt="Hoodie" onerror="this.src='https://via.placeholder.com/300'">
      <h3>Winter Hoodie</h3>
      <p>99 TND</p>
   </div>
</section>

<footer>
   &copy; 2025 BoutiqueMode.tn – All rights reserved.
</footer>

</body>
</html>
EOF'
```

STEP 1: Upload images to S3 (PRIVATE)





STEP 2: Confirm EC2 has the **default learner IAM role**



STEP 3: Test S3 access again

```
[ec2-user@ip-172-31-19-112 ~]$ aws s3 ls s3://boutiquemode-tn-website-ys-mbn

Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-172-31-19-112 ~]$ ^C
[ec2-user@ip-172-31-19-112 ~]$ aws s3 ls s3://boutiquemode-tn-website-ys-mbn
2025-12-06 18:22:49       4502 bag.jpg
2025-12-06 18:22:50      17935 dress.jpg
2025-12-06 18:22:51       8361 hero.jpg
2025-12-06 18:22:51      15574 hoodie.jpg
2025-12-06 18:23:00     688324 shoes.jpg
[ec2-user@ip-172-31-19-112 ~]$
```

Before assigning EC2 to a Lab Role

After Attaching the Role

STEP 4: Download all images from S3 to EC2

aws s3 cp s3://boutiquemode-tn-website-ys-mbn/ /home/ec2-user/ --recursive

```
[ec2-user@ip-172-31-19-112 ~]$ aws s3 cp s3://boutiquemode-tn-website-ys-mbn/ /home/ec2-user/ --recursive
download: s3://boutiquemode-tn-website-ys-mbn/hoodie.jpg to ./hoodie.jpg
download: s3://boutiquemode-tn-website-ys-mbn/bag.jpg to ./bag.jpg
download: s3://boutiquemode-tn-website-ys-mbn/dress.jpg to ./dress.jpg
download: s3://boutiquemode-tn-website-ys-mbn/hero.jpg to ./hero.jpg
download: s3://boutiquemode-tn-website-ys-mbn/shoes.jpg to ./shoes.jpg
[ec2-user@ip-172-31-19-112 ~]$
```

STEP 5: Move the images into Nginx's public folder

sudo mv /home/ec2-user/*.jpg /usr/share/nginx/html/

sudo systemctl restart nginx

```
[ec2-user@ip-172-31-19-112 ~]$ sudo mv /home/ec2-user/*.jpg /usr/share/nginx/html/
[ec2-user@ip-172-31-19-112 ~]$ sudo chmod 644 /usr/share/nginx/html/*.jpg
[ec2-user@ip-172-31-19-112 ~]$ sudo systemctl restart nginx
[ec2-user@ip-172-31-19-112 ~]$ ls -lha /usr/share/nginx/html/
total 764K
drwxr-xr-x. 3 root     root      16K Dec  6 18:34 .
drwxr-xr-x. 4 root     root       33 Dec  6 18:09 ..
-rw-r--r--. 1 root     root     3.6K Aug 12 21:18 404.html
-rw-r--r--. 1 root     root     3.7K Aug 12 21:18 50x.html
-rw-r--r--. 1 ec2-user ec2-user 4.4K Dec  6 18:22 bag.jpg
-rw-r--r--. 1 ec2-user ec2-user  18K Dec  6 18:22 dress.jpg
-rw-r--r--. 1 ec2-user ec2-user 8.2K Dec  6 18:22 hero.jpg
-rw-r--r--. 1 ec2-user ec2-user  16K Dec  6 18:22 hoodie.jpg
drwxr-xr-x. 2 root     root       27 Dec  6 18:09 icons
-rw-r--r--. 1 root     root     2.7K Dec  6 18:14 index.html
-rw-r--r--. 1 root     root      368 Aug 12 21:18 nginx-logo.png
lrwxrwxrwx. 1 root     root       14 Aug 12 21:19 poweredby.png -> nginx-logo.png
-rw-r--r--. 1 ec2-user ec2-user 673K Dec  6 18:23 shoes.jpg
[ec2-user@ip-172-31-19-112 ~]$
```

S3 + EC2 + Nginx OK.